Mobile Application

Modes of Operations:

1) OFF State: EN1, EN2, IN1, IN2, IN3, IN4 are all set to zero.
2) Balance in Place: EN1, EN2, IN1, IN2, IN3, IN4 are constantly changed to reach a balance state.
3) Move and Balance:
   - **Move Left and Balance:** In order to move left, the robot has to make a 90° counterclockwise rotation around its vertical axis, then move forward. This can be done by fixing the left wheel and rotate the right wheel for some time until the robot makes 90°.
   - **Move Right and Balance:** In order to move right, the robot has to make a 90° clockwise rotation around its vertical axis, then move forward. This can be done by fixing the right wheel and rotate the left wheel for some time until the robot makes 90°.
   - **Move Forward and Balance:** In order to move forward, the wheels have to rotate in the same direction while maintaining a balanced mode.
   - **Move Backward and Balance:** In order to move backward, the wheels have to rotate in the same direction but opposite to the current direction while maintaining a balanced mode.

The robot will decide which action to take according to the received message from the Bluetooth module. This message will be one of the following: $00x, 10x, 11x, 12x, 13x, 14x$. The ‘x’ character at the end of the message is a delimiter character to indicate that the message has ended.

Action Decoding:

DIGIT 1	DIGIT 2	ACTION
0	0	OFF
1	0	Balance in Place
1	1	Move Left and Balance
1	2	Move Right and Balance
1	3	Move Forward and Balance
1	4	Move Backward and Balance